Design Thinking Report – AI-Powered Travel Planner

Step 1: Empathize – Understanding User Needs

User Research Summary:

Research drew insights from travel forums, interviews, and mobile-first surveys. It aimed to uncover user pain points related to using AI-generated travel planning tools, especially on mobile devices.

Key Insights:

- Users feel overwhelmed by too many auto-suggested options.
- AI-generated itineraries feel too general and impersonal.
- Editing itineraries is slow and requires multiple steps.
- Mobile users want quicker, on-the-go editing tools.
- Users experience visual fatigue due to poor lighting adaptability.
- Some users with mild visual impairments struggle with contrast, font sizes, or layout clutter.

Typical User Persona – Anabel Jones:

Anabel, 29, is a Lagos-based content creator. She frequently travels for work and leisure and plans trips on her phone. While she appreciates AI-generated ideas, she finds it difficult to make quick customizations and is sensitive to screen brightness and interface clutter when planning at night or outdoors.

Step 2: Define – Framing the Problem

Problem Statement:

“Mobile users of the AI-powered travel planner often abandon their planning sessions midway due to overwhelming suggestions, limited personalization control, and lack of accessible visual and functional support.”

Design Goals:

- Simplify the customization of AI itineraries on mobile.
- Allow instant control over activities without disrupting flow.
- Reduce cognitive load with clean, minimalist layouts.
- Improve accessibility through visual theme settings, adjustable text, and screen reader compatibility.

- Support inclusivity with built-in accessibility preferences.

Step 3: Ideate – Generate Possible Solutions

Brainstormed Ideas:

- Drag-and-Drop Timeline Editing – Rearranging activities visually.
- Quick Swap Button – Instantly replace items with smart alternatives.
- Mood Filter Toggle – Regenerate plans by theme (e.g., Relaxing, Cultural).
- Voice Command Editing – Use speech to modify plans.
- Mini Preview Popups – View activity info without leaving the screen.
- Dark/Light Mode Toggle – Adapt screen contrast for lighting conditions.
- Accessibility Settings Panel – Include options to:
    - Adjust font sizes and contrast
    - Enable screen reader-friendly layout
    - Turn on simplified mode for users with cognitive fatigue

Selected Concept:

A Drag-and-Drop Timeline Editor with a Quick Swap Button, paired with a Dark/Light Mode Toggle and a dedicated Accessibility Settings menu located under Profile > App Settings. This enhances customization, comfort, and inclusivity for mobile users.

Step 4: Reflect – Process, Challenges & Next Steps

Design Process Summary:

This Design Thinking journey involved deeply understanding how users interact with AI-generated plans and what causes frustration or disengagement. We uncovered that both functional control and visual adaptability are key to user satisfaction.

Challenges Faced:

- Balancing AI-driven automation with freedom to edit.
- Designing for diverse user needs without overwhelming mobile layouts.
- Including accessible features while keeping settings easy to find and use.
- Addressing both visual and interaction-based accessibility needs.

Plan for Moving Forward:

- Design and prototype:
    - Drag-and-Drop Timeline Interface
    - Quick Swap Button
    - Dark/Light Mode Toggle
    - Accessibility Panel (Font Size, Contrast, Simplified UI)
- Conduct usability testing with a focus group (5–7 mobile-first users, including at least one with visual sensitivity)
- Iterate based on accessibility feedback
- Prepare handoff specs for development integration